

How to write your own article by latex with AI

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Why use $\text{\LaTeX}$ for articles

- High-quality typesetting (math, tables, figures)
- Stable cross-references and bibliographies
- Reproducible workflows (scripts, CI)
- Works well with AI-assisted writing tools for drafting and polishing



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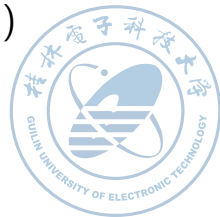
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Recommended tools

- Distribution: TinyTeX or TeX Live
- Engines: XeLaTeX or LuaLaTeX (better font support)
- Build: latexmk (automates multiple runs)
- Editor: VSCode + LaTeX Workshop, or Overleaf (cloud)



Common package installation

Typical install command (TeX Live / TinyTeX):

```
1 tlmgr install ctex biblatex algorithm2e  
2 listings xstring biber latexindent
```

Build example:

```
1 latexmk -xelatex main.tex  
2 biber main  
3 latexmk -xelatex main.tex
```



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Recommended article structure

- ① Abstract
- ② Introduction
- ③ Related Work
- ④ Method / Theory
- ⑤ Experimental Design
- ⑥ Results & Analysis
- ⑦ Discussion
- ⑧ Conclusion
- ⑨ References & Appendix



Practical sectioning tips

- Draft an outline first
- Keep one main idea per section
- Use labels (`\label{...}`) and `\ref` for cross-references
- Put long code and large tables in appendix or external repo



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Math typesetting best practices

- Use `amsmath` environments: `align`, `gather`
- Number only referenced equations
- Define macros for repeated notation (vectors, operators)



Algorithm example

Algorithm 1: Example optimization pseudocode

Input: initial parameter x_0 , tolerance ϵ

Output: optimized parameter x^*

initialize $B_0 \leftarrow I$;

while $\|\nabla f(x_t)\| > \epsilon$ **do**

$p_t \leftarrow -B_t \nabla f(x_t)$;

 perform line search to get α_t ;

$x_{t+1} \leftarrow x_t + \alpha_t p_t$;

$s_t \leftarrow x_{t+1} - x_t$, $y_t \leftarrow \nabla f(x_{t+1}) - \nabla f(x_t)$;

 update B_{t+1} by BFGS formula;



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Code management

- Keep code in a `code/` directory; reference or link in the paper
- For short snippets use `listings` inline; avoid pasting very long scripts
- Provide a requirements file and clear reproduction steps



Code snippet example

```
1 def gradient_step(w, grad, lr):  
2     return w - lr * grad  
3  
4 for epoch in range(epochs):  
5     grad = compute_gradient(w)  
6     w = gradient_step(w, grad, lr)
```



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Using biblatex + biber

- Preamble:
 - `\usepackage[backend=biber,style=authoryear]{biblatex}`
 - `\addbibresource{references.bib}`
- Compile process: `xelatex biber xelatex` &2
- If biber missing: install and add to PATH



Fallback: manual references

If automated tools unavailable, list references manually in an appendix. Prefer .bib management for research papers.



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Useful automation

- **latexmk** –automated compilation
- **biber** –bibliography backend
- **latexindent** –code/tex formatter
- CI (GitHub Actions) –reproducible builds



Common environment issues

- `biber`: command not found install biber and add to PATH
- `latexindent` not found install and configure editor path
- Missing `.sty` files `tlmgr install <pkg>`



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Recommended workflow

- ① Organize content into sections/ files
- ② Keep figures in figures/, code in code/
- ③ Use a build script or Makefile
- ④ Push to repo, use CI to verify build



Example build script

```
1 #!/bin/bash
2 set -e
3 latexmk -xelatex -interaction=nonstopmode main.tex
4 biber main || true
5 latexmk -xelatex -interaction=nonstopmode main.tex
6 latexmk -c
```



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Pre-submission checklist

- All references compile and appear correctly
- Figures have captions and are referenced
- Tables use booktabs, no vertical rules
- Code and data are accessible or archived
- Spelling and grammar checked



Final tips

- Start from a minimal working example
- Use macros for repeated notation
- Script builds; avoid manual multiple-step builds
- Keep the project folder organized



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Further reading

- TeX Stack Exchange: <https://tex.stackexchange.com>
- TinyTeX / TeX Live documentation
- LaTeX Workshop for VSCode



WELCOME FEEDBACK!

